project a growth rate of 0.5% by 2050, which would add 49 million more people per year if the population climbs to a projected 9.8 billion. The reduction in annual growth rate observed over the past five decades shows that the human population is now growing more slowly than expected in exponential growth. This change resulted from fundamental shifts in population dynamics due to diseases, including AIDS, and to voluntary population control.

Regional Patterns of Population Change

We have described changes in the global population, but population dynamics vary widely from region to region. In a stable regional population, birth rate equals death rate (disregarding the effects of immigration and emigration). Two possible configurations for a stable population are

$$\text{Zero population growth} = \text{High birth rate} - \text{High death rate}$$

or

$$\text{Zero population growth} = \text{Low birth rate} - \text{Low death rate}$$

The movement from high birth and death rates toward low birth and death rates, which tends to accompany industrialization and improved living conditions, is called the **demographic transition**. In Sweden, this transition took about 150 years, from 1810 to 1975, when birth rates finally approached death rates. In Mexico, where the human population is still growing rapidly, the transition is projected to take until at least 2050. Demographic transition is associated with an increase in the quality of health care and sanitation as well as improved access to education, especially for women.

After 1950, death rates declined rapidly in most developing countries, but birth rates have declined more variably. Birth rates have fallen most dramatically in China. The expected number of children per woman per lifetime in China decreased from 5.9 in 1970 to 1.6 in 2011, largely due to the government's strict "one-child" policy. In some countries of Africa, the transition to lower birth rates has also been rapid, though birth rates remain high in most of sub-Saharan Africa.

How do such variable birth rates affect the growth of the world's population? In industrialized nations, populations are near equilibrium, with reproductive rates near the replacement level of 2.1 children per female (over their lifetime). In many industrialized countries—including the United States, Canada, Germany, Japan, and the United Kingdom—total reproductive rates are in fact *below* the replacement level. These populations will eventually decline if there is no immigration and if the birth rate does not change. In fact, the population is already declining in many eastern and central European countries. Most of the current global population growth occurs in less industrialized countries, where about 80% of the world's people now live.

A unique feature of human population growth is our ability to control family sizes through planning and voluntary contraception. Social change and the rising educational and career aspirations of women in many cultures encourage women to delay marriage and postpone reproduction. Delayed reproduction helps to decrease population growth rates and to move a society toward zero population growth under conditions of low birth rates and low death rates. However, there is a great deal of disagreement as to how much support should be provided for global family planning efforts.

Age Structure

Another important factor that can affect population growth is a country's **age structure**, the relative number of individuals of each age in the population. Age structure is commonly graphed as "pyramids" like those in **Figure 53.23**. For Zambia, the pyramid is bottom heavy, skewed toward young individuals who will grow up and perhaps sustain the explosive growth with their own reproduction. The age structure for the United States is relatively even until the older, postreproductive ages. Although the current total reproductive rate in the United States is about 1.8 children per woman—below the replacement rate—the population is projected to grow slowly through 2050 as a result of immigration. For Italy, the pyramid has a small base, indicating that individuals younger than reproductive age are relatively underrepresented in the population. This situation contributes to the projection of a future population decrease in Italy.

Age-structure diagrams not only predict a population's growth trends but also can illuminate social conditions. Based on the diagrams in Figure 53.23, we can predict that employment and education opportunities will continue to be a problem for Zambia in the foreseeable future. In the United States and Italy, a decreasing proportion of younger working-age people will soon be supporting an increasing population of retired "boomers." This demographic feature has made the future of Social Security and Medicare a major political issue in the United States. Understanding age structures is necessary to plan for the future.

Infant Mortality and Life Expectancy

Infant mortality, the number of infant deaths per 1,000 live births, and *life expectancy at birth*, the predicted average length of life at birth, vary widely in different countries. In 2015, for example, the infant mortality rate was 71 (7.1%) in Angola but only 2 (0.2%) in Japan. Life expectancy at birth was 59 years in Angola but 85 years in Japan. These differences reflect the quality of life faced by children at birth and influence the reproductive choices parents make. If infant